

Supplementary Material: Response-Aware Marginal-Utility Routing for Multi-Context Prediction

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Abstract

This supplement documents the controlled environment, the implementation and training configuration of R-MUR, the multi-target traffic protocol, the complete per-seed results, and the additional response-observability analysis that supports the mechanism section of the main paper.

1. Controlled Environment

1.1. Data generator

Each episode contains a target vector $y \in \mathbb{R}^G$, an anchor prediction $a = y + \sigma_a \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, I)$, and K candidate contexts. The candidate pool is built from $U = \max(1, \text{round}(K(1 - r_{\text{dup}})))$ distinct groups, where r_{dup} is the duplicate ratio. Each candidate is assigned to one of these groups; duplicates share the same group assignment. For a candidate in group g , the useful measurement is $m_g = y_g + \rho_{\text{use}}(a_g - y_g)$ with $\rho_{\text{use}} = 0.1$. A candidate marked harmful receives $m_g = y_g + \rho_{\text{harm}}(a_g - y_g)$ with $\rho_{\text{harm}} = 1.8$. The frozen expert predicts each group by the mean measurement of its selected candidates, leaving groups without a selected candidate at their anchor value. The episode loss is the mean squared error across groups. Standalone and state-conditioned marginals are computed by enumerating every candidate addition. Relevance is drawn independently from $[0.7, 1.0]$; duplicate groups therefore make content redundancy observable to similarity-based selectors. Maximum marginal relevance follows the standard greedy rule with relevance minus a cosine-similarity penalty to the selected set; the penalty is selected on a validation split from $\{0, .25, .5, 1\}$.

Table 1: Controlled protocols. All reported gains are loss reductions relative to the anchor-only predictor. Seeds are independent training and evaluation runs.

Protocol	K	B	r_{dup}	Seeds / splits
Redundancy ladder	16	4	0, .25, .5, .75	5 seeds; 3000 train, 2500 test
Candidate scaling	4, 8, 16, 32, 64	4	.50	5 seeds; 2000 train, 1500 test

1.2. Protocols

The controlled experiments use 15 training epochs for the candidate-scaling protocol and 20 epochs for the redundancy ladder.

2. R-MUR Implementation and Training

2.1. Router

The router has three parts: a cached scorer g_C , a response-aware scorer g_R , and one shared set encoder. Every candidate is represented by its history, the query by the anchor history, and the selected set by the histories of the selected candidates packed into a fixed-size tensor with a validity mask. A permutation-invariant DeepSets encoder of depth two with hidden and set width 64 summarises the selected set. The cached scorer maps the query, set and candidate representations to a scalar. The response-aware scorer takes the same inputs plus the seven response features described below and is a two-layer network of width 64.

2.2. Response summary

For a candidate j at state A the frozen predictor is evaluated with and without the candidate, and the summary concatenates: the mean of the base forecast, the mean of the forecast after adding the candidate, the mean of the change, its standard deviation, its mean absolute value, its Euclidean norm divided by $\sqrt{H}$ with horizon H , and its maximum absolute value.

2.3. Training

Training states are sampled from the training episodes. A state is a uniformly random subset of the candidate pool whose size is drawn uniformly from 0 to $B - 1$, so empty, partial and nearly complete selected sets all occur. Every candidate outside the state contributes one labelled example with its true marginal utility, computed as the counterfactual loss difference between

the predictor with and without that candidate. A single state therefore supplies $K - |A|$ regression targets and all candidate pairs of the state for the ranking term. The objective is a Huber regression term plus a pairwise ranking term whose per-pair weight is the absolute utility gap of the pair,

$$\mathcal{L} = \mathcal{L}_{\text{Huber}} + \beta \sum_{i,j: \tilde{m}_i > \tilde{m}_j} |\tilde{m}_i - \tilde{m}_j| \log(1 + \exp[-(\hat{m}_i - \hat{m}_j)]), \quad (1)$$

with $\beta = 1$ in every reported run. Training uses AdamW with learning rate 10^{-3} and weight decay 10^{-4} , batches of 128 states, and 20 epochs; each target-run contributes 16,000 states, four per training episode, with 16 candidate labels each.

2.4. Inference

At deployment the cached scorer ranks the remaining candidates and the response-aware scorer is evaluated only on the top- q shortlist, with $q = 4$ in every reported run. The router adds the highest-scoring shortlisted candidate when its estimated marginal utility is positive and otherwise stops; it also stops when the context budget is reached. The predictor is therefore evaluated on a fraction q/K of the pool per decision.

2.5. Baselines

Standalone Utility is a model with the same set encoder, trained on states with an empty selected set and one sampled candidate per state, so it estimates the standalone marginal $u(j \mid \emptyset)$ and its ranking is fixed before selection. The cached-only router is the same model trained on random nonempty states from cached inputs only, so it is state-conditioned but never observes the predictor response. Both use the same optimiser, schedule and state count as R-MUR.

Content-based subset rules. Facility location maximises the sum of similarities to the target, greedy DPP maximises the determinant of the candidate kernel, facility location with relevance weights multiplies the two, and k -center minimises the largest distance from the target to the selected set. All four use the same candidate pool and budget and require no training.

Fixed-set acquisition rules. Five policies from the feature-selection and active-feature-acquisition literature are adapted to the protocol. Every policy uses only the training and validation splits: the harm rate is the per-episode fraction of episodes whose loss exceeds the anchor-only loss, the same definition used for every other policy in the paper. Each chooses one set per target from the training and validation splits with the frozen predictor, and then applies that set to every test episode; none of them observes a test label, and none conditions its decision on the state of an individual episode. They answer a question the router must answer, namely whether per-episode conditioning buys anything over the best set that can be chosen offline.

- *Greedy validation set* follows best-subset and optimal-pursuit selection [?]: starting from the empty set, it adds the candidate with the largest validation loss reduction, and one elimination pass then removes the weakest element and reconsiders its best replacement. The elimination pass selected the same set as plain forward selection on all 576 target-runs of the sweep, so the two are not reported separately.
- *Adaptive-k relevance* follows adaptive- k context selection [?]: candidates are ordered by relevance and the number of contexts is chosen on validation rather than fixed in advance, so the policy may select fewer than the budget when additional contexts do not reduce the validation loss.
- *Cluster-based selection* follows iterative demonstration selection [?]: the candidate pool is clustered in history space and the most relevant member of each cluster represents it, so the chosen set is diverse by construction and needs no predictor. The cluster structure is estimated on the validation split; no statistic of the test episodes enters the selection, and the fixed set is then applied to them.
- *Two-step lookahead set* follows nongreedy active feature acquisition [? ?]: the best pair of candidates is chosen on validation before any single candidate is committed to, and the set is then completed greedily, so jointly informative candidates are not missed by a myopic rule. The policy selects the same set as greedy validation selection on most runs and matches its gain to within .0002 on every dataset, so non-greedy lookahead does not change the outcome in this setting.

- *Gradient influence* ranks candidates by the first-order loss reduction, so the sign convention matters: $2\langle d_j, r \rangle + \|d_j\|^2$ is the increase in loss and the ranking uses its negation. The policy is scored on the validation split and the resulting set is applied to the test episodes.
- *Gradient influence* follows gradient-estimated demonstration selection [?]: the first-order term of the squared-loss change, $2\langle f_A - y, d_j \rangle + \|d_j\|^2$ with $d_j = f_{A \cup \{j\}} - f_A$, scores each candidate on the validation split, averaged over randomly sampled states so that the score does not depend on one state; candidates are scored independently, in linear time in the pool size. Because that expression is a loss increase, the ranking selects the candidates with the smallest value.

The oracle policies use the true marginals and are not deployable.

3. Extensions of the Regret Decomposition

The main paper states the shortlist-aware bound

$$m(j^* \mid A) - m(\hat{j} \mid A) \leq 2\epsilon_C + 2\epsilon_R, \quad (2)$$

which follows from the exact decomposition into a screening and a reranking term. One remark extends it.

Sequential selection.. A guarantee for a sequence of B selections requires the induced set utility to be monotone and submodular, so that approximate greedy selection inherits the standard $(1 - 1/e)$ factor up to the accumulated estimation error. Context utility is not monotone in general, because a context can increase the loss, and we do not assume submodularity in the experiments. The sequential case is therefore stated as a condition rather than claimed for R-MUR.

4. Multi-Target Traffic Protocol

The six benchmarks use the standard files distributed with the traffic forecasting benchmark suite. Each file provides a multivariate series and fixed train, validation, and test index sets; no observation is reordered across the split boundaries. Values are standardized with the mean and standard deviation of the training segment.

For a dataset with N sensors, the evaluation fixes 32 evenly spaced target sensors. Each target sensor has its own candidate pool, built from the training segment: eight sensors with the highest correlation to the target, five sensors of intermediate correlation, and three sensors drawn uniformly at random with a fixed construction seed. Each episode takes a window from the split under evaluation. The anchor is the target sensor’s previous 12 observations, each candidate contributes its previous 12 observations, and the target is the target sensor’s next 12 observations.

The frozen predictor is a ridge model with penalty 10, fitted for each target sensor on that target’s training windows under the full context subset, the empty subset, and a randomly masked subset of its own candidate pool, and then held fixed for candidate evaluation, oracle computation and router training. The predictor is therefore specific to the target and is refitted for every target. The loss is the mean squared error over the forecast horizon.

Each target is repeated with three independent runs. The reported prediction gain is the reduction in loss relative to the anchor-only predictor, the negative-transfer rate is the fraction of episodes whose routed loss exceeds the anchor-only loss, and oracle recovery is the pooled ratio of learned gain to sequential-oracle gain. Intervals resample the 32 target means with 5000 bootstrap replicates; the hierarchical variant resamples targets and then runs within each target.

5. Complete Controlled and Multi-Target Results

6. Additional Response-Observability Analysis

The probe behind the mechanism panel of the main paper uses the frozen ridge predictor of a single target sensor of METR-LA and five independent runs. It samples four states per episode. For each state and candidate the feature families are:

- $\mathcal{X}_1$: query history, mean selected history, candidate history, their difference and product, and selected-set size.
- $\mathcal{X}_2$: $\mathcal{X}_1$ plus the mean expert prediction before and after adding the candidate, the mean prediction delta, and the mean absolute delta.
- $\mathcal{X}_3$: $\mathcal{X}_2$ plus the full before and after forecast trajectories and their difference.

A histogram-based gradient boosting regressor and classifier are fitted for each family, with hyperparameters selected on a calibration split over depth $\{3, 6, \infty\}$ and L_2 regularization $\{10^{-3}, 1\}$, 150 boosting iterations and learning rate .08. The classifier provides the positive-utility AUC; the regressor provides the mean absolute error, Spearman correlation, top-1 accuracy and one-step regret reported in the main paper. The probe is a diagnostic of the information content of the predictor response, not a routing policy.

7. Decision-Margin Audit

The candidate-scaling audit uses the saved rollout predictions and true marginals of the controlled scaling protocol. For each state and episode, let $m_{(1)}$ and $m_{(2)}$ be the largest and second-largest true marginal utilities among the unselected candidates. The positive top-two margin is $\Delta = m_{(1)} - m_{(2)}$ when $\Delta > 0$. We report the normalized pointwise error

$$\text{nMAE} = \frac{\mathbb{E}|\hat{m} - m|}{\text{SD}(m)}, \quad (3)$$

the fraction of states with zero top-two margin, the median ratio between the maximum absolute candidate error and the positive margin, and the decision accuracy

$$\text{Acc}_{\text{dec}} = \Pr\left[m(\hat{j} \mid A) = m(j^* \mid A)\right], \quad (4)$$

where $\hat{j}$ is the predicted argmax and j^* is a true argmax. Strict top-1 accuracy compares candidate indices, so it counts an exact tie among equally useful candidates as an error. Accuracy and margin statistics use states with at least two unselected candidates, where a nontrivial argmax decision exists.

Table 2: Per-seed controlled redundancy ladder at $K = 16$ and $B = 4$. Gains are loss reductions relative to the anchor-only predictor.

r_{dup}	Seed	Oracle-Static	R-MUR	Oracle-Greedy	H_{state}
0	101	.691	.663	.691	.000
0	202	.678	.655	.678	.000
0	303	.697	.677	.697	.000
0	404	.692	.669	.692	.000
0	505	.681	.653	.681	.000
0.25	101	.517	.502	.585	.115
0.25	202	.509	.489	.574	.113
0.25	303	.518	.510	.584	.114
0.25	404	.523	.516	.588	.111
0.25	505	.513	.497	.575	.108
0.5	101	.353	.369	.450	.216
0.5	202	.347	.361	.442	.215
0.5	303	.353	.366	.449	.213
0.5	404	.350	.366	.451	.224
0.5	505	.353	.364	.444	.204
0.75	101	.176	.206	.247	.286
0.75	202	.175	.205	.244	.283
0.75	303	.179	.208	.250	.282
0.75	404	.179	.217	.253	.293
0.75	505	.176	.207	.246	.285

Table 3: Per-seed candidate-count scaling under fixed redundancy. Normalized MAE divides pointwise error by the standard deviation of the true marginal targets. Decision accuracy counts a selection as correct when it attains the best true utility, including exact ties.

K	Seed	R-MUR recovery	Normalized MAE	Positive margin	Strict top-1	Decision accuracy	One-step regret
4	101	.961	.387	.205	.470	.873	.0192
4	202	.960	.412	.230	.493	.887	.0164
4	303	.969	.426	.216	.491	.890	.0167
4	404	.979	.382	.230	.495	.887	.0201
4	505	.969	.408	.221	.492	.883	.0186
8	101	.799	.612	.116	.227	.383	.0654
8	202	.809	.565	.111	.237	.403	.0573
8	303	.800	.613	.113	.228	.378	.0666
8	404	.843	.556	.112	.252	.416	.0535
8	505	.815	.574	.116	.240	.396	.0608
16	101	.687	.698	.071	.174	.285	.0693
16	202	.699	.681	.066	.183	.298	.0656
16	303	.709	.687	.071	.185	.301	.0650
16	404	.708	.679	.072	.182	.299	.0667
16	505	.684	.684	.071	.176	.289	.0687
32	101	.604	.751	.042	.141	.224	.0608
32	202	.600	.733	.041	.141	.220	.0588
32	303	.591	.718	.044	.140	.222	.0618
32	404	.586	.728	.044	.137	.222	.0626
32	505	.598	.728	.040	.140	.219	.0586
64	101	.470	.718	.025	.089	.136	.0513
64	202	.484	.731	.025	.098	.151	.0482
64	303	.479	.756	.023	.096	.155	.0479
64	404	.486	.739	.022	.088	.149	.0484
64	505	.480	.751	.023	.094	.146	.0483

Table 4: Per-seed METR-LA observability probe. $\mathcal{X}_1$ uses cached context features, $\mathcal{X}_2$ adds scalar response summaries, and $\mathcal{X}_3$ adds full forecast trajectories. MAE is the mean absolute error and AUC is the area under the receiver operating characteristic curve.

Seed	Input	MAE	Spearman	Positive AUC	Top-1	Regret
101	$\mathcal{X}_1$	.0339	.169	.651	.173	.0635
101	$\mathcal{X}_2$	.0308	.377	.829	.260	.0503
101	$\mathcal{X}_3$	.0307	.333	.839	.242	.0505
202	$\mathcal{X}_1$	.0400	.162	.649	.158	.0677
202	$\mathcal{X}_2$	.0351	.383	.838	.208	.0538
202	$\mathcal{X}_3$	.0353	.383	.843	.237	.0531
303	$\mathcal{X}_1$	.0418	.144	.655	.124	.0796
303	$\mathcal{X}_2$	.0370	.391	.848	.225	.0580
303	$\mathcal{X}_3$	.0376	.363	.855	.223	.0623
404	$\mathcal{X}_1$	.0405	.172	.640	.131	.0807
404	$\mathcal{X}_2$	.0371	.390	.828	.285	.0667
404	$\mathcal{X}_3$	.0370	.391	.835	.262	.0654
505	$\mathcal{X}_1$	.0427	.171	.652	.154	.0765
505	$\mathcal{X}_2$	.0384	.411	.827	.320	.0559
505	$\mathcal{X}_3$	.0384	.393	.834	.258	.0563

Table 5: Structural headroom per dataset over 32 fixed targets and five runs (960 target-run evaluations). Every evaluation has positive headroom.

Dataset	Mean headroom	Median headroom	Positive fraction
METRLA	.097	.088	1.000
PEMSBAY	.115	.116	1.000
PEMS03	.122	.111	1.000
PEMS04	.085	.076	1.000
PEMS07	.089	.087	1.000
PEMS08	.078	.077	1.000

Table 6: Per-seed multi-target prediction gains for the learned policies, averaged over the 32 fixed targets of each dataset.

Dataset	Run	Standalone	Cached-only	R-MUR
METRLA	101	.001	.002	.015
METRLA	202	.001	.006	.012
METRLA	303	.005	.005	.017
PEMSBAY	101	.023	.027	.055
PEMSBAY	202	.027	.028	.052
PEMSBAY	303	.031	.030	.060
PEMS03	101	.015	.015	.027
PEMS03	202	.016	.017	.029
PEMS03	303	.014	.014	.025
PEMS04	101	.024	.025	.035
PEMS04	202	.025	.024	.035
PEMS04	303	.025	.025	.037
PEMS07	101	.014	.014	.028
PEMS07	202	.017	.016	.029
PEMS07	303	.014	.015	.028
PEMS08	101	.025	.025	.034
PEMS08	202	.024	.024	.033
PEMS08	303	.024	.025	.033

Table 7: Content-based subset-selection baselines under the multi-target protocol, averaged over 32 fixed targets and three runs. None of these rules trains a model or evaluates the predictor during selection.

Dataset	Facility location	DPP greedy	Facility loc. + rel.	k -center
METRLA	-.021	-.015	-.017	-.018
PEMSBAY	.008	.005	-.007	-.014
PEMS03	.009	.007	-.002	.004
PEMS04	.014	.013	.012	.012
PEMS07	.007	.006	-.002	.003
PEMS08	.017	.012	.013	.009
Mean	.006	.005	-.000	-.000